THE LAWS OF SOFTWARE TEAMS

Hofstadter's law

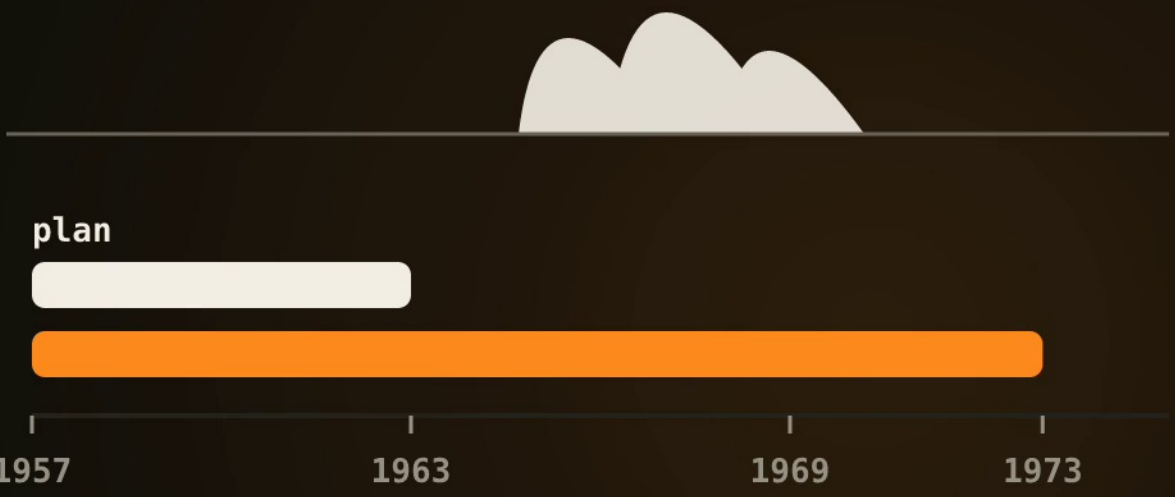
It always takes longer than you expect, even when you plan for it

swipe →

Ivan Rykov

Sydney Opera House

Planned: 1963
Opened: 1973



cost **$102M**

10 years late · ~15× the budget

02 THE LAW

“It always takes longer than you expect, even when you take into account Hofstadter’s law”



re-estimates: 3

Always ten years away

Douglas Hofstadter · Gödel, Escher, Bach · 1979

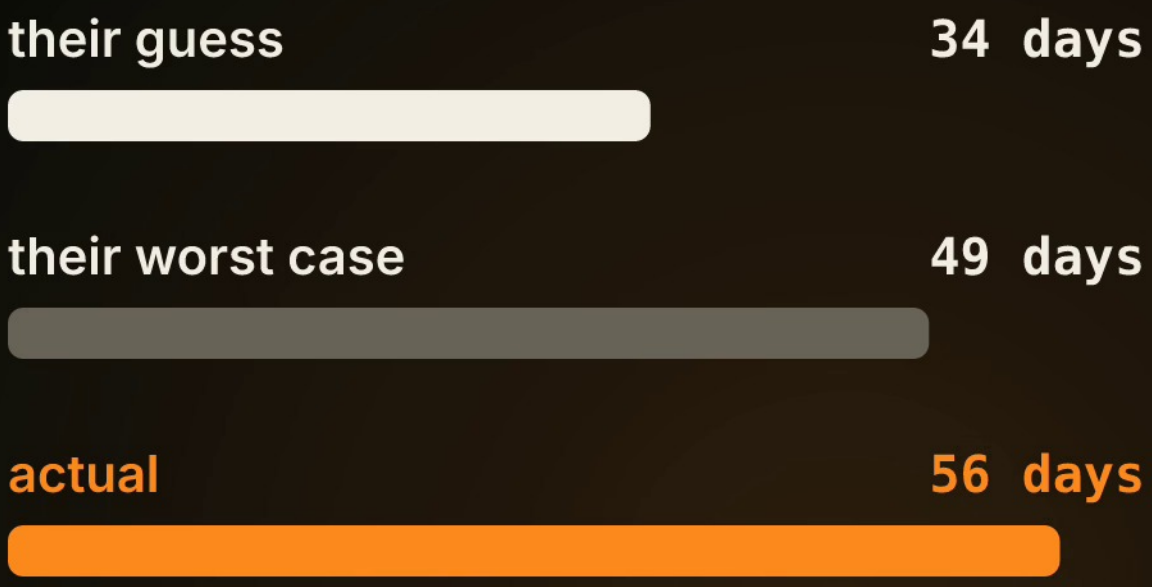
When will a computer beat the chess champion?



1997: Deep Blue beats Kasparov

Even the worst case was too optimistic

Students predicting when they'd finish a thesis



Only 30% finished by their own guess

Buehler, Griffin & Ross, 1994 · the "planning fallacy"

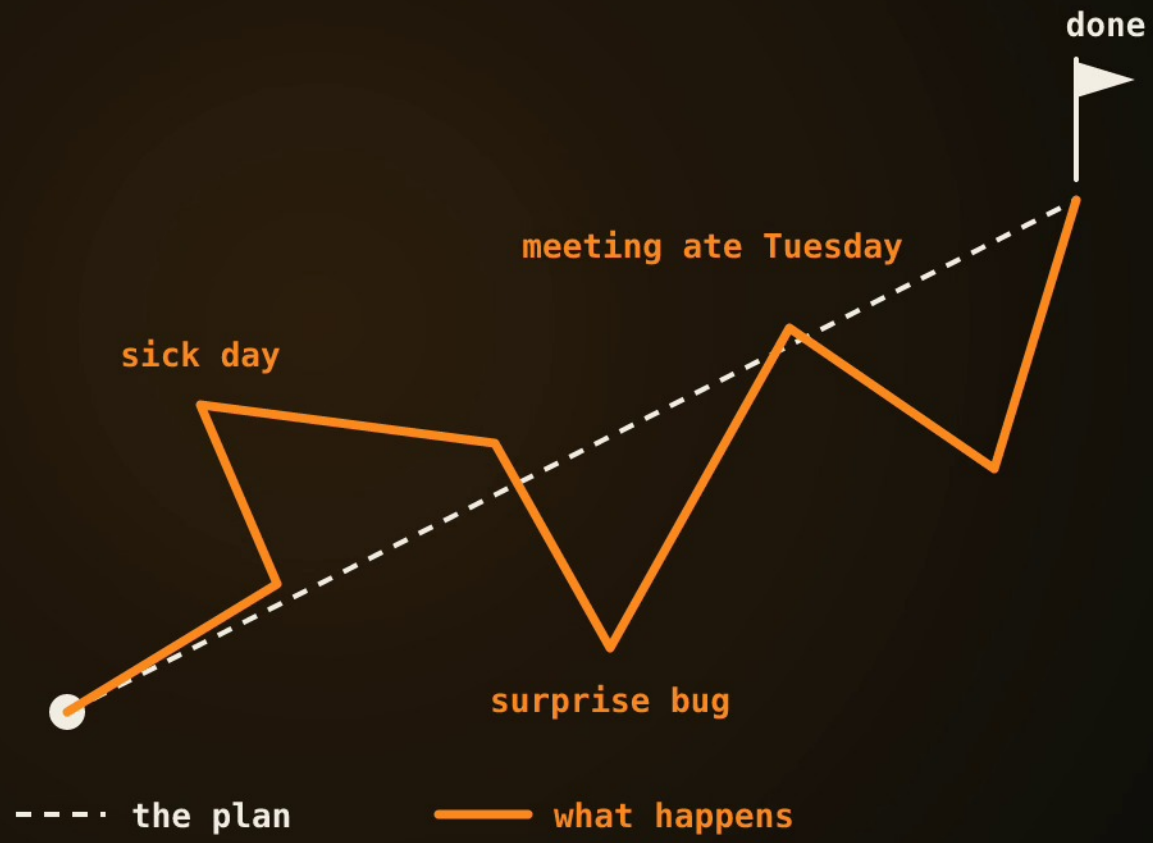
1 in 6 IT projects blows up



+200% cost **+70%** time

1,471 IT projects · Flyvbjerg & Budzier, 2011 · grid illustrative

We plan the best case



Reality takes the scenic route

Don’t ask how long
Ask how long it took

similar past projects · days



“Reference class forecasting”

Kahneman & Tversky · data illustrative

PARKINSON

Don't pad

HOFSTADTER

You'll be late anyway

Plan with your history, not your hopes

TAKEAWAY

Plan with your history, not your hopes

Sources: D. Hofstadter, Gödel, Escher, Bach, 1979 · Buehler, Griffin and Ross, 1994 · Flyvbjerg and Budzier, 2011

next: the 90-90 rule